SHAIRA AFRA REDHILA

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RESEARCH INTERESTS

Biomaterials and Bio-nanomaterials; Hydrogel-based Drug Delivery and Wound Healing; Tissue Engineering and Regenerative Medicine; Bio-technology and Bio-nanotechnology; Biosensors and Biomedical Devices; Computational Biology.

EDUCATION

Khulna University of Engineering & Technology *CGPA: 3.84/4.00 (Ranked 4th in overall merit position)*
Bachelor of Science in Biomedical Engineering *March, 2022- September, 2026*

- **UG Thesis**: "Development and Optimization of Physically Cross-linked Chitosan/PVA/Gelatin Composite Hydrogels for Antibiotic Loaded Wound Dressing Applications".
- Developed and optimized chitosan/PVA/gelatin composite hydrogels using freeze-thaw physical crosslinking, investigating the effects of polymer stock concentrations and mixture ratios on wound-dressing performance.
- Conducted physicochemical, mechanical, thermal, morphological, and preliminary drug-release and antibacterial characterization, including swelling, water retention, gel fraction, porosity, biodegradation, WVTR, FTIR, UTM, TGA-DSC, UV-Vis spectroscopy, and antibacterial testing.

PUBLICATIONS

Conference Publications

1. S. B. Shuvo, S. Hossain**, S. A. Redhila**, and U. Bose, "Exploring Common Molecular Interactions across Multiple Cancers to Identify Potential Therapeutic Targets and Drug Candidates," EICT 2025, KUET.
2. S. Datta, G. M. M. Rahman, P. Majumder, **S. A. Redhila**, et al., "A Voice-Assisted Smart Device for the Visually Impaired People: Leveraging AI Vision, Object Detection, and GPS-Based Navigation," IEEE BECITHCON 2025.

Manuscripts in Preparation

3. **S. A. Redhila**, S. Tasnim, K. N. Hasan, & K. Z. Ferdwushee, "Development and Optimization of Physically Cross-linked Chitosan/PVA/Gelatin Composite Hydrogels for Antibiotic-Loaded Wound Dressing Applications."
4. K. N. Hasan, **S. A. Redhila**, & G. M. M. Rahman, "Integrated transcriptomic analysis for host-directed target discovery in measles virus infection."
5. S. B. Shuvo, **S. A. Redhila**, S. Hossain, & A. D. Roy, "NeuroSwin: A Swin Transformer-GRU Model for Parkinson's Disease Classification and Brain Region Localization from EEG Signals." Manuscript in preparation for Biomedical Signal Processing and Control.

TECHNICAL SKILLS

- **Polymer Processing:** Hydrogel preparation, freeze-thaw crosslinking, freeze-drying, drug loading and release.
- **Physicochemical Characterization:** Swelling, water retention, gel fraction, biodegradation, porosity, pH responsiveness, and water vapor transmission rate (WVTR).
- **Instrumental Characterization:** Fourier-transform infrared spectroscopy (FTIR), UV-Vis spectroscopy, thermogravimetric analysis (TGA), differential scanning calorimetry (DSC), scanning electron microscopy (SEM), and X-ray diffraction (XRD).
- **Mechanical & Biological Testing:** Universal testing machine (UTM), Brine Shrimp Lethality Assay test, antimicrobial testing, Blood Clotting Index measurement.
- **Computational & Simulation:** Python; MATLAB; COMSOL Multiphysics; OriginPro; Ansys.
- **Data Science & Bioinformatics:** Bulk and single-cell RNA-seq analysis; differential expression; functional enrichment; protein-protein interaction networks.
- **Others:** Latex, MS Word, MS PowerPoint, MS Excel, Canva

PROJECTS

AI-Driven Glucose Monitoring System for Telemedicine Application | *BME 4232, KUET* | May 2026

- Designed a low-cost spectrophotometric glucose as an alternative to costly lab equipment for resource-constrained telemedicine use

Temperature-Regulated CPAP System for Sleep Apnea Management | *BME 3200, KUET* | Feb-May 2024

- Engineered a thermistor-based CPAP prototype that auto-adjusts air pressure based on breathing-cycle temperature changes

Arterial Wall Mechanics | *BME 3114, KUET* | Feb-May 2024

- Built a COMSOL computational model of arterial walls using the Holzapfel-Gasser-Ogden (HGO) framework to simulate hyperelastic and viscoelastic behavior of multi-layer, fiber-reinforced tissue

Implementation of a Digital Hearing Aid System using MATLAB | *BME 3112, KUET* | May 2024

- Simulated a full digital hearing-aid signal chain: noise injection, wavelet vs. lowpass denoising, FFT-based frequency-shaping/gain modification, and amplitude normalization

Implementation of an Artificial Pacemaker | *BME 2232, KUET* | Oct-Nov 2023

- Designed and simulated a basic pacemaker circuit to detect abnormal ECG rhythms and deliver corrective electrical pulses, demonstrating feasibility of low-cost cardiac support systems

TRAININGS AND WORKSHOPS

- **Training on Laboratory Animals Used in Biomedical Sciences | icddr,b, 23-25 Aug 2026 |** Hands-on training in the handling and care of laboratory animals, including mice, rats, guinea pigs, and rabbits; Practical exposure to animal handling, routes of administration, animal husbandry, and breeding facility operations; Introduction to wound induction procedures and their relevance to wound-healing research.
- **Industrial Attachment in Biomedical Engineering |** National Electro-Medical Equipment Maintenance Workshop & Training Center and Technical College of Allied Health | Specialized government hospitals | 2026
- **Technical Workshop on Biomaterials-Based Wet Lab | BUET, Oct 2025 |** Gained practical exposure to biomaterials-based wet-lab techniques and experimental procedures relevant to biomaterials research.
- Data Analysis with Python (80 hrs) | CCC, KUET (EDGE Project, Bangladesh Computer Council | Nov–Dec 2024
- Technical workshop on Research Design & Thesis Writing using LaTeX | Advanced Bio-Engineering Club, KUET
- Technical workshop on Crafting Innovation-Entrepreneurship & Hardware in Fab Lab Ecosystem | KUET, Feb 2024

CERTIFICATIONS

- AI for Medical Diagnosis | DeepLearning.AI (Coursera) | Jun 2025
- Supervised Machine Learning: Regression and Classification | *DeepLearning.AI* | May 2025
- Secondary 2 Upper Intermediate (English) | *British Council* | 2018

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **KUET Career Club** | Vice President (Sep 2025–Sep 2026); Senior Executive, IT & Resources (Jun 2024–Sep 2025); Junior Executive, IT & Resources (Mar 2023–Jun 2024) | IT & Designs, Public Relations
- Junior Training Coordinator, **Advanced Bio-Engineering Club of KUET** | Apr 2023 to Sep 2026
- Campus Ambassador, **Turtle Venture** | Jan 2024 to Jan 2025
- **Champion**, Darts (Female) and **Organizer**, BME Indoor 4.0, Biomedical Engineering Association, KUET | 2025

HONORS & AWARDS

- **Dean's Awards:** GPA 3.98/4.00,4th year; GPA 3.89/4.00, 3rd year, September 2026; GPA 3.83/4.00, 1st year, September 2023 | Provided by Faculty of Electrical and Electronic Engineering for outstanding academic performance
- **Executive award:** Best Performer for three consecutive years, KUET Career Club.
- **Scholarships:** Talentpool Scholarship in the Primary School Certificate (PSC) Examination, awarded by the Board of Primary Education, Dhaka; General Scholarships in the Secondary School Certificate (SSC) and Higher Secondary Certificate (HSC) Examinations, awarded by the Board of Secondary and Intermediate Education, Dhaka.

REFERENCES

- Dr. Md. Bashir Uddin, Professor, (Project Supervisor), Department of Biomedical Engineering, Khulna University of Engineering & Technology (KUET) | Contact: *mbu@bme.kuet.ac.bd*
- Kazi Zannatul Ferdowshee, (Thesis Supervisor), Assistant Professor, Department of Biomedical Engineering, Khulna University of Engineering & Technology (KUET) | Contact: *ferdwushee@bme.kuet.ac.bd*